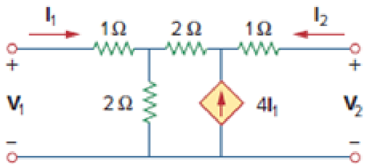
Chapter 19, Problem 82P

Problem

Use *PSpice* or *MultiSim* to rework Prob. 19.31.

Determine the hybrid parameters for the network in Fig. 19.89.

Figure 19.89



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

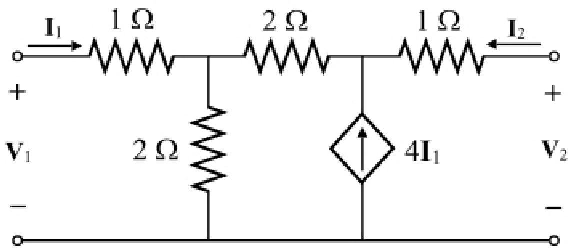


Figure 1

Step 2 of 9

Write the hybrid parameters.

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} \quad \ldots\ldots (1)$$

$$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0} \quad \ldots\ldots (2)$$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} \quad \ldots\ldots (3)$$

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0} \quad \ldots\ldots (4)$$

Step 3 of 9

PSpice analysis:

• Open the CIS Demo software.

• Once you are in the capture session frame, click on the menu item File, select new, and then click on Project.

• In the New Project box, type 'selected file name' in the Name text box. Ensure that the Analog or Mixed-signal circuit Wizard is activated.

• Connected the circuit diagram as per the Circuit.

• Double click on each component and enter the given values.

• Click the New Simulation Profile icon and enter a name in the Name text box. You will need to enter the appropriate settings for this project in the Simulation

• Setting box. Click the Analysis tab, and select Bias Point from the Analysis type list. Click OK and save the document.

• Click on the run icon. You will observe the A/D Demo screen. When you close this screen, you will observe the bias voltages and currents.

Step 4 of 9

To calculate the values of h_{11} and h_{21} parameters, connect a 1 V test voltage source to input port and short circuit the output port.

The following is the PSpice schematic:

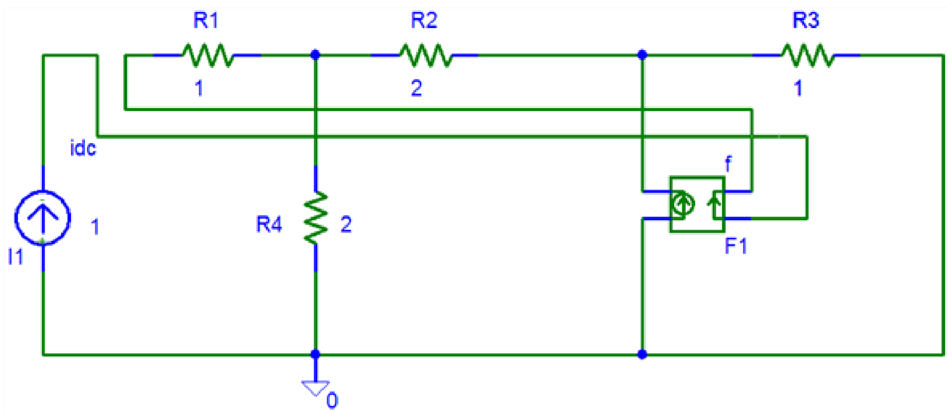


Figure 2

Step 5 of 9

Run the Schematic, the following is the schematic circuit with voltage and current values.

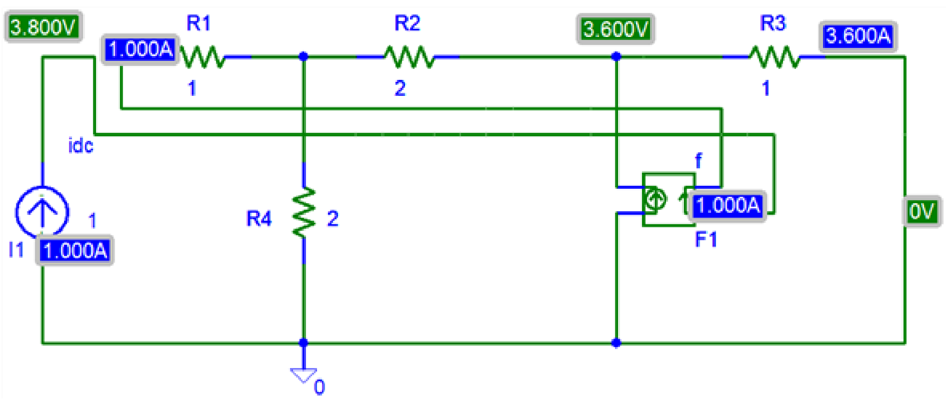


Figure 3

Step 6 of 9

From the Figure 3,

$\mathbf{I}_1 = 1\text{ A}$
$\mathbf{I}_2 = -3.6\text{ A}$
$\mathbf{V}_1 = 3.8\text{ V}$
$\mathbf{V}_2 = 0\text{ V}$

Substitute 3.8 V for $\mathbf{V}_1$ in the equation (1).

$$\begin{aligned}\mathbf{h}_{11} &= \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}\\ &= \frac{3.8}{1}\\ &= 3.8\,\Omega\end{aligned}$$

Substitute 1 A for $\mathbf{I}_1$ and -3.6 A for $\mathbf{I}_2$ in the equation (3).

$$\begin{aligned}\mathbf{h}_{21} &= \left.\frac{\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}\\ &= \frac{-3.6}{1}\\ &= -3.6\end{aligned}$$

Step 7 of 9

To calculate the values of $\mathbf{h}_{12}$ and $\mathbf{h}_{22}$ parameters, connect a 1 V test current source to output port and open circuit the input port.

The following is the PSpice schematic:

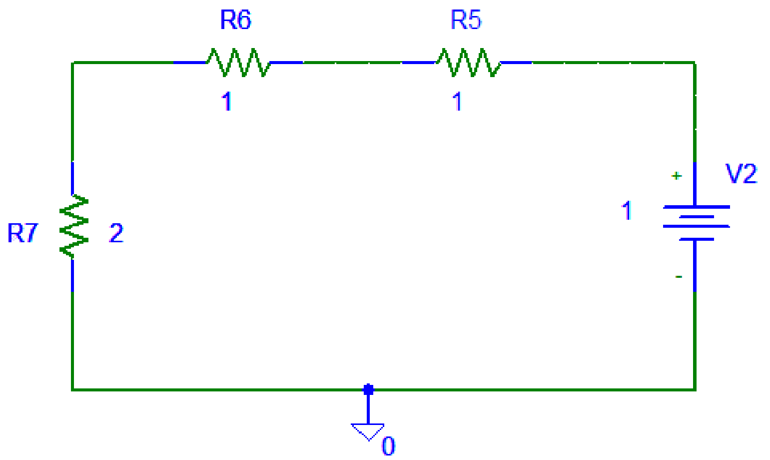


Figure 4

Step 8 of 9

Run the Schematic, the following is the schematic circuit with voltage and current values.

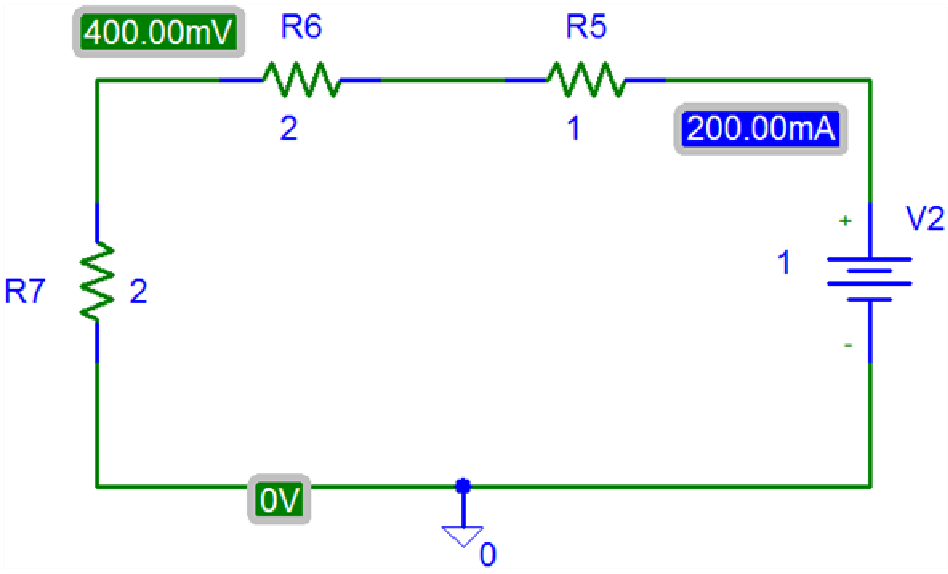


Figure 5

Step 9 of 9

From the Figure 5,

$$\mathbf{I}_1 = 0\ \mathrm{A}$$
$$\mathbf{I}_2 = 200\ \mathrm{mA}$$
$$\mathbf{V}_1 = 400\ \mathrm{mV}$$
$$\mathbf{V}_2 = 1\ \mathrm{V}$$

Substitute $400\ \mathrm{mV}$ for $\mathbf{V}_1$ and $1\ \mathrm{V}$ for $\mathbf{V}_2$ in the equation (2).

$$\begin{aligned}\mathbf{h}_{12} &= \left.\frac{\mathbf{V}_1}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0} \\ &= \frac{400\times10^{-3}}{1} \\ &= 0.4\end{aligned}$$

Substitute $200\ \mathrm{mA}$ for $\mathbf{I}_2$ and $1\ \mathrm{V}$ for $\mathbf{V}_2$ in the equation (4).

$$\begin{aligned}\mathbf{h}_{22} &= \left.\frac{\mathbf{I}_2}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0} \\ &= \frac{200\ \mathrm{mA}}{1\ \mathrm{V}} \\ &= 0.2\ \mathrm{S}\end{aligned}$$

Therefore, the hybrid parameters, $\begin{bmatrix}\mathbf{h}_{11} & \mathbf{h}_{12}\\ \mathbf{h}_{21} & \mathbf{h}_{22}\end{bmatrix}$ are $\boxed{\begin{bmatrix}3.8\ \Omega & 0.4\\ -3.6 & 0.2\ \mathrm{S}\end{bmatrix}}$.